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Hossein Sharifzadeh

SUMMARY

Senior Staff Engineer, Lead Solutions Architect, and AI Platform Engineer with 8+ years of experience designing and operating large-scale distributed data platforms, cloud-native architectures, real-time streaming systems, and AI-powered applications. Proven track record leading enterprise-scale initiatives involving Apache Spark, Kafka, HDFS, Databricks, cloud platforms, and advanced data architectures supporting mission-critical workloads.

Trusted advisor to executive leadership with extensive experience presenting technical strategy, platform roadmaps, AI initiatives, and modernization programs to CIOs, CEOs, and senior stakeholders. Currently focused on AI Agents, enterprise AI architecture, TOGAF, and scalable platform engineering for next-generation intelligent systems.

SKILLS

Leadership & Strategy: Team Leadership, Stakeholder Engagement, Technical Roadmaps, Executive Presentations, Cross-Functional Collaboration, TOGAF Framework, Agile, Scrum

Languages: Python, Scala, Java, SQL, PL/SQL, Shell scripting, C++.

Big Data: Apache Spark, Kafka, Hive, HDFS, Sqoop, Trino.

Cloud & DevOps: AWS, OCI, Azure, Docker, Linux, Shell scripting

Data Management & ETL: Data Modeling, Erwin, PowerDesigner, ODI, SSIS, OBIEE, Databricks.

Databases: Oracle, PostgreSQL, Redshift, Hive, Snowflake.

AI & Research: Azure AI, open source LLMs, PGVector, RAG, Open-WebUI, Langchain, HuggingFace.

EXPERIENCE

Royal Bank of Canada, Canada— *Senior Staff Engineer/Lead Solutions Architect*

2025 – present

- Lead architecture and engineering initiatives for large-scale compliance and surveillance platforms processing high-volume financial data across distributed systems.
- Architect migration of mission-critical compliance datasets from kdb+/KX technology to Spark and **HDFS**-based distributed platforms, improving scalability, resiliency, and operational efficiency.

- Designed and deployed AI-powered solutions leveraging **LLMs, RAG** architectures, vector databases, semantic retrieval techniques, and agentic workflows to automate compliance investigations, intelligent document generation, and trade-surveillance analysis.
- Developed scalable data processing pipelines using **Spark, Python, Scala, Databricks**, and cloud-native engineering practices supporting enterprise-grade reliability and performance requirements.
- Established engineering standards covering CI/CD, automated testing, code quality, observability, deployment automation, and operational excellence.
- Conduct architecture reviews, mentor engineers, and provide technical leadership across multiple teams while collaborating directly with senior business stakeholders.
- Drive evaluation and adoption of emerging AI platform technologies, LLM architectures, and enterprise AI governance practices.

Atlantic Lottery Corporation, Canada – *Senior Data Engineer & Solutions Architect*

2022 – 2025

- Designed and built scalable **data pipelines** supporting real-time and batch analytics across departments.
- Led data **infrastructure strategy**, including modeling standards, metadata tooling, and platform modernization.
- Developed **AI agents** using **RAG** and **LLMs** to assist non-technical users in querying the data warehouse.
- Reported directly to the **CIO** and consulted **architectural** decisions and project outcomes to C-suite **stakeholders**.
- Initiated and drove enterprise-level solution evaluations for semantic modeling, AI-assisted querying, and **cloud optimization**.
- Mentored engineers and analysts; fostered cross-team innovation through architecture reviews and working groups.

Toolset: ODI(Oracle Data Integrator, similar to AWS Glue and Apache Airflow), Jira, SSIS, Oracle database, SQL Server, Oracle Analytics, Power BI, Bash scripting, Python scripting, AWS Solutions Architect, OCI Solutions Architect, Open-WebUI.

Huawei Technologies & Dalhousie University, Canada – *Research Assistant*

Aug 2021 – Aug 2023

- Creating a simulation for Fiber optics channels.
- **Inventing** a new state-of-the-art Encoder and Decoder system to enhance the performance of **optical fiber** channels.
- Managed project milestones, coordinated deliverables with Huawei teams, and presented progress to research sponsors.
- Trained junior researchers and organized collaborative technical reviews.

Toolset: Matlab coding, Shell scripting, HPC clusters.

BRTech *Big Data Solutions Architect*

Mar 2020 – Aug 2021

- Designing systems like **active archiving**, **real-time data stream** processing, and providing client services and APIs.
- designing data pipelines and creating ETL applications using **Apache Spark**.
- Directed migration plan of 2PB of telecom archive data from Oracle to HDFS using Spark and Kafka.
- Led cross-functional teams in the design and deployment of distributed data pipelines.
- Reported directly to the **CEO**; presented solution architectures, cost models, and research-backed strategies.
- Evaluated emerging **big data solutions** and authored internal technical papers for decision-making at the executive level.

Toolset: Apache Spark (code in Scala), Apache Hive, Impala, Sqoop, ELK (Elastic, Logstash, Kibana), Grafana, other tools from Cloudera stack (HDFS, Kafka, Zookeeper, etc.), Shell scripting, Python.

BRTech – *DW/BI Senior Engineer*

APR 2019 – Feb 2021

- L2 and L3 of the support for resolving and debugging daily **ETL** jobs.
- Supervised the **BI development team**; managed DW operations, data integration, and dashboard/report delivery.
- Collaborated with department heads to gather requirements and align deliverables with business objectives.
- Create and maintain the Data warehouse **data models**.
- Designing the L1 check manual, using SQL and **PL-SQL** scripts.
- Creating a test environment from scratch to mimic the Prod for developers to test their code.

Toolset: ODI, OBIEE, Oracle Database, Oracle toad, SQL Developer, Shell Scripting, Erwin data modeler, SAP powerdesigner, Python.

Parsian Robotics Lab – *AI & Embedded Systems Supervisor*

May 2017 – Apr 2020

- Supervised a multidisciplinary team working on AI algorithms, embedded systems, and robotics hardware.
- Led the integration of speech recognition systems like CMU PocketSphinx for vocal command on autonomous robots, and managed the development of FPGA-based communication protocols with error detection over UART.
- Oversaw PCB design using Altium Designer and embedded programming in Verilog and VHDL.
- Provided technical mentorship, coordinated between hardware and AI teams, and ensured project alignment with research goals, all while collaborating closely with academic advisors and contributing to national robotics competitions.

EDUCATION

Dalhousie University, Halifax, Canada – *Master of Applied Science in Electrical and Computer Engineering*

2021 – 2023

Research: Simulation and improve the performance of the Optical Fiber Networks

Supervisor: Prof. Dmitry Trukhachev

RESEARCH & TECHNICAL PROJECTS

AI Agent for SQL Query Generation (LLM + RAG) (2025)

- Developed AI assistant tool that translates business questions into SQL queries for self-service analytics.

Data Warehouse Cloud Infrastructure Strategy Enhancements(2024)

- Cross-cloud architecture initiatives by evaluating and comparing Oracle Cloud (OCI), AWS, and Azure to identify the **most efficient** and cost-effective **solution** for enterprise **data warehousing**.
- Took on a **Solutions Architect** role to assess compute performance, network efficiency, and platform-native services, resulting in optimized resource utilization, automated scaling strategies, and improved system performance.
- Implemented architectural best practices to align cloud usage with business goals and presented findings directly to C-suite leadership and higher managers.

Secure Cloud Systems via Homomorphic Encryption (2020)

- Built a Python-based system for computation over encrypted data to secure cloud-based processing.

Telecom Data Lake Migration & SPARK ETL (2020)

- Led architecture and implementation of Spark ETL pipelines that consumed Kafka topics and inserted data into Hive.

RoboCup Vision and Navigation System (2019)

- Implemented vision and pathfinding algorithms for RoboChallenge League in FIRA World Cup (South Korea).

Certifications and Training

AWS Certified Solutions Architect – Associate – AWS

Dec 2024 – Dec 2027

Certificate for Exercising Leadership: Foundational Principles – HBS

Mar 2025

Oracle Cloud Infrastructure 2024 Certified Foundations Associate – Oracle

Dec 2024

DevOps Professional Certificate – PagerDuty